TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Dry Test for Acid Radicals

Preliminary examination of a salt — physical observations, heating in a dry tube, and the action of dilute and concentrated sulphuric acid: the first evidence for the acid radical.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 1 · LEVEL 1 TERM 1

1. Aim and place of the experiment

Aim: to identify the acid radical (anion) present in a given salt by dry (preliminary) tests. Qualitative analysis begins with dry tests because they cost no reagents, take minutes, and narrow the field: their evidence is *indicative, not final* — every radical suggested here must be confirmed by a wet test (Experiment 3) before it is reported. The golden rule of the sessional: **observe everything, infer cautiously, confirm always.**

2. Step 1 — Physical examination

OBSERVATION	INDICATION
Colour: blue / blue-green	Copper salts (Cu^{2+})
Colour: green	Nickel or ferrous salts, chromates' partner cations
Colour: pink / violet	Manganese or cobalt salts
Colour: yellow / brown	Ferric salts, chromates
Colourless / white	None of the coloured cations; common salts of Na^+, K^+, Ca^{2+}, Zn^{2+}, Al^{3+}, NH_4^+…
Smell of vinegar on rubbing / warming	Acetate (CH_3COO^-)
Smell of rotten eggs	Sulphide (S^{2-})
Smell of ammonia (moisten and warm gently; test with red litmus or HCl rod)	Ammonium salt (NH_4^+)
Deliquescent (turns wet in air)	Chlorides/nitrates of Ca, Mg, Zn, Fe commonly

3. Step 2 — Heating in a dry test tube

Heat a little salt gently, then strongly, in a perfectly dry tube; note decrepitation (crackling), colour changes, sublimate and gas:

OBSERVATION ON HEATING	INDICATION
White sublimate on the cooler part of the tube	Ammonium salt, or mercury compounds
Yellow sublimate	Sulphur compounds / some iodides
Brown fumes (NO_2)	Nitrate or nitrite present
Crackling (decrepitation)	NaCl, KCl, $Pb(NO_3)_2$ — crystals with enclosed moisture
Colourless gas relighting a glowing splint (O_2)	Nitrate of an alkali metal, or chlorate
Residue yellow when hot, white when cold	Zinc compound (ZnO behaviour — a classic)
Residue blackens (charring)	Organic radical — acetate, oxalate, tartrate

4. Step 3 — Action of dilute H_2SO_4 (cold, then warm)

The workhorse test: add dilute sulphuric acid to a little salt; gases evolved from the acid radicals announce themselves:

OBSERVATION	GAS	RADICAL INDICATED
Brisk effervescence; gas turns lime water milky	CO_2	Carbonate, CO_3^{2-}
Rotten-egg smell; gas blackens lead-acetate paper	H_2S	Sulphide, S^{2-}
Burning-sulphur smell; gas turns acidified dichromate paper green	SO_2	Sulphite, SO_3^{2-}
Brown fumes, increasing on warming	NO_2	Nitrite, NO_2^-
Vinegar smell on warming	CH_3COOH vapour	Acetate, CH_3COO^-
No action with dilute acid	—	Carbonate/sulphide/sulphite/nitrite/acetate absent; proceed to concentrated acid

5. Step 4 — Action of concentrated H_2SO_4 (warm)

OBSERVATION	RADICAL INDICATED
Colourless pungent gas; dense white fumes with a rod dipped in NH_4OH (NH_4Cl)	Chloride, Cl^- (gas is HCl)
Reddish-brown vapour, intensified by adding copper turnings	Bromide, Br^- (vapour is Br_2), or nitrate (see next line — distinguish by copper: bromine colour deepens uniformly; nitrate gives deep brown NO_2 especially on heating with Cu)
Violet vapour, condensing to grey-black solid	Iodide, I^- (vapour is I_2)
Brown fumes, greatly increased by copper turnings or paper pellets	Nitrate, NO_3^-
Effervescence; gas burns with a blue flame (CO) when the acid acts on the dry salt	Oxalate (CO + CO_2 evolved)
Charring (blackening) of the salt	Organic matter — acetate/tartrate/sugar-like radical

Safety, stated once for the whole course: concentrated H_2SO_4 is added in small quantity, warmed gently, and the tube is never pointed at anyone; gases are smelled by wafting, never by direct inhalation; H_2S and Br_2 tests are done in the fume cupboard where available.

6. Recording the result

Sessional records are written in the three-column discipline — **Test | Observation | Inference** — one line per test, observations in plain factual words ("colourless gas, rotten-egg smell; lead-acetate paper blackened") and inferences modest ("sulphide indicated; to be confirmed"). The examiner marks the discipline as much as the answer: an inference written as a fact before its confirmatory test costs marks every time.

7. Precautions

- The tube for the heating test must be perfectly dry, or steam imitates reactions.
- Use small quantities — dry tests are preliminary; the salt must last the whole analysis.
- Add acid down the side of the tube to avoid spurting; warm gradually.
- Never taste; smell by wafting; treat every unknown as poisonous until identified.

8. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. A salt gives brisk effervescence with dilute H_2SO_4 and the gas turns lime water milky. Name the gas and the radical, and the confirmatory wet test you will run in Experiment 3.
2. Distinguish sulphite from carbonate when both effervesce with dilute acid.
3. A white salt gives a white sublimate on heating and a smell of ammonia when warmed with NaOH. Which two constituents are indicated?
4. With concentrated H_2SO_4 a salt gives dense white fumes with an NH_4OH rod but no coloured vapour. Name the gas and radical.
5. Why is the dry test called preliminary, and what must happen before the radical is reported as found?
6. State the observation identifying an oxalate with concentrated H_2SO_4 and the property of one of its gases used in the test.
7. A salt crackles on heating and its residue is yellow when hot but white when cold. Report both observations with their textbook indications.

Answer key

1. Gas: CO_2; radical: carbonate (CO_3^{2-}). Confirmation (wet): the soda-extract test — the extract acidified gives effervescence, and $CaCl_2$ gives a white precipitate of $CaCO_3$ dissolving in acid with effervescence.
2. Pass the gas over acidified dichromate paper/solution: SO_2 from a sulphite turns it green; CO_2 from a carbonate has no effect. (Also: sulphite's gas smells of burning sulphur; carbonate's is odourless.)
3. An ammonium salt (sublimate and ammonia with alkali both indicate NH_4^+ — a basic radical found by dry tests as a bonus).
4. HCl gas; chloride (Cl^-). Confirmation to follow: $AgNO_3$ test (Experiment 3).
5. Because several radicals can give similar indications and mixtures mask one another, dry tests only suggest candidates; a specific wet confirmatory test must be positive before the radical is reported.
6. Effervescence with evolution of CO_2 and CO; the CO burns with a blue flame when ignited at the tube's mouth — the property used.
7. Crackling (decrepitation): crystals holding enclosed moisture — e.g. NaCl, KCl, $Pb(NO_3)_2$. Yellow-hot/white-cold residue: zinc oxide behaviour — a zinc salt indicated.